COHERENCE OF THE PUSHFORWARD OF THE STRUCTURE SHEAF UNDER A DIMENSION CONDITION ON THE ASSOCIATED PRIMES OF THE COMPLETE LOCAL RINGS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-local-cohomology/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The reduction of the coherence of a pushforward to a local statement, and the description of the pushforward through the cohomology of the complement. Both are invoked at the start of the counted proof (source lines 1846-1850) as pointers to lemmas of this chapter. Reduce to a Noetherian local domain with the distinguished point in the complement.
- The extension of a section over a smaller open to the whole scheme after multiplication by a power of an element vanishing on the complement, and the criterion for coherence of a pushforward under a depth condition. Invoked inside the counted proof (source lines 1859-1863 and 1884-1885) as cross-chapter pointers and a pointer to a lemma of this chapter. Reduce further to the case where the complement is the closed point, and then transfer the statement from the completion.

Source reference. Original lemma lemma-check-finiteness-local-cohomology-locally: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 1847–1847. Navigation pointer only: the counted text refers to the Stacks item lemma-check-finiteness-local-cohomology-locally, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-finiteness-pushforwards-and-H1-local: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 1848–1848. Navigation pointer only: the counted text refers to the Stacks item lemma-finiteness-pushforwards-and-H1-local, which belongs to another chapter of the project and is not part of this

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Source reference. Original lemma lemma-finiteness-pushforward: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 1884–1884. Navigation pointer only: the counted text refers to the Stacks item lemma-finiteness-pushforward, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference coherent-lemma-homs-over-open: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 1680–1680. Navigation pointer only: the counted text refers to the Stacks item coherent-lemma-homs-over-open, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference algebra-remark-Noetherian-complete-local-ring-universally-catenary: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 1882–1882. Navigation pointer only: the counted text refers to the Stacks item algebra-remark-Noetherian-complete-local-ring-universally-catenary, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference algebra-lemma-Noetherian-complete-local-Nagata: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 1883–1883. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-Noetherian-complete-local-Nagata, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference coherent-lemma-flat-base-change-cohomology: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 1892–1892. Navigation pointer only: the counted text refers to the Stacks item coherent-lemma-flat-base-change-cohomology, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference descent-lemma-finite-type-descends: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 1895–1895. Navigation pointer only: the counted text refers to the Stacks item descent-lemma-finite-type-descends, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let X be an integral locally Noetherian scheme. Let $j : U \to X$ be the inclusion of a nonempty open subscheme with complement Z. Assume that for all $z \in Z$ and any associated prime $\mathfrak{p}$ of the completion $\mathcal{O}^\wedge_{X,z}$ we have $\dim(\mathcal{O}^\wedge_{X,z}/\mathfrak{p}) \geq 2$. Then $j_*\mathcal{O}_U$ is coherent.*

Proof. We may assume X is affine. Using Lemmas lemma-check-finiteness-local-cohomology-locally and lemma-finiteness-pushforwards-and-H1-local we reduce to $X = \mathrm{Spec}(A)$ where $(A, \mathfrak{m})$ is a Noetherian local domain and $\mathfrak{m} \in Z$. Then we can use induction on $d = \dim(A)$. (The base case is $d = 0, 1$ which do not happen by our assumption on the local rings.) Set $V = \mathrm{Spec}(A) \setminus \{\mathfrak{m}\}$. Observe that the local rings of V have dimension strictly smaller than d. Repeating the arguments for $j' : U \to V$ we and using induction we conclude that $j'_*\mathcal{O}_U$ is a coherent $\mathcal{O}_V$-module. Pick a nonzero $f \in A$ which vanishes on Z. Since $D(f) \cap V \subset U$ we find an n such that multiplication by f^n on U extends to a map $f^n : j'_*\mathcal{O}_U \to \mathcal{O}_V$ over V (for example by Cohomology of Schemes, Lemma coherent-lemma-homs-over-open). This map is injective hence there is an injective map

$$j_*\mathcal{O}_U = j''_*j'_*\mathcal{O}_U \to j''_*\mathcal{O}_V$$

on X where $j'' : V \to X$ is the inclusion morphism. Hence it suffices to show that $j''_*\mathcal{O}_V$ is coherent. In other words, we may assume that X is the spectrum of a local Noetherian domain and that Z consists of the closed point.

Assume $X = \mathrm{Spec}(A)$ with $(A, \mathfrak{m})$ local and $Z = \{\mathfrak{m}\}$. Let $A^\wedge$ be the completion of A. Set $X^\wedge = \mathrm{Spec}(A^\wedge)$, $Z^\wedge = \{\mathfrak{m}^\wedge\}$, $U^\wedge = X^\wedge \setminus Z^\wedge$, and $\mathcal{F}^\wedge = \mathcal{O}_{U^\wedge}$. The ring $A^\wedge$ is universally catenary and Nagata (Algebra, Remark algebra-remark-Noetherian-complete-local-ring-universally-catenary and Lemma algebra-lemma-Noetherian-complete-local-Nagata). Moreover, condition (3) of Lemma lemma-finiteness-pushforward for $X^\wedge, Z^\wedge, U^\wedge, \mathcal{F}^\wedge$ holds by assumption! Thus we see that $(U^\wedge \to X^\wedge)_*\mathcal{O}_{U^\wedge}$ is coherent. Since the morphism $c : X^\wedge \to X$ is flat we conclude that the pullback of $j_*\mathcal{O}_U$ is $(U^\wedge \to X^\wedge)_*\mathcal{O}_{U^\wedge}$ (Cohomology of Schemes, Lemma coherent-lemma-flat-base-change-cohomology). Finally, since c is faithfully flat we conclude that $j_*\mathcal{O}_U$ is coherent by Descent, Lemma descent-lemma-finite-type-descends. $\square$